RECOVERY OF SPARSE SIGNALS

A. FIXTURE

1. Setting

Let A be an $n \times m$ measurement matrix with $n < m$, and consider the program

$$\min_x \|x\|_{\ell_1} \quad \text{subject to} \quad Ax = y. \tag{1}$$

Program (1) is known as basis pursuit [CDS98]. Its stability was studied in [CRT06, Theorem 1].

$$(1 - \delta_S)\|c\|^2 \le \|A_T c\|^2 \le (1 + \delta_S)\|c\|^2. \tag{2}$$

The inequality (2) is the restricted isometry property [CT05].

2. Main result

Theorem 2.1. *If $\delta_{3S} + 3\delta_{4S} < 2$, then the solution of (1) recovers every S-sparse signal.*

Proof. Combine (2) with Lemma 2.2 below. □

Lemma 2.2. *The error $h = x^\sharp - x_0$ obeys the cone constraint $\|h_{T^c}\|_{\ell_1} \le \|h_T\|_{\ell_1}$.*

Proof. This follows from the optimality of $x^\sharp$ in (1); see also [Don06]. □

References

[CDS98] S. S. Chen, D. L. Donoho and M. A. Saunders, Atomic decomposition by basis pursuit, SIAM J. Sci. Comput. 20 (1998), 33–61.

[CRT06] E. J. Candès, J. Romberg and T. Tao, Stable signal recovery from incomplete and inaccurate measurements, Comm. Pure Appl. Math. 59 (2006), 1207–1223.

[CT05] E. J. Candès and T. Tao, Decoding by linear programming, IEEE Trans. Inform. Theory 51 (2005), 4203–4215.

[Don06] D. L. Donoho, Compressed sensing, IEEE Trans. Inform. Theory 52 (2006), 1289–1306.